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DSA0404-Fundamentals of Data Science

1.Detecting an Unusual Customer

Aim

To apply the concept of the z-score to determine how unusual a customer's spending is compared to the average spending of all customers.

Problem Statement

A shopping website has an average customer spending of ₹2,000 with a standard deviation of ₹400. One customer spends ₹3,200. Calculate the customer's z-score, determine whether the spending is unusually high, explain the meaning of a negative z-score, and interpret another customer's z-score of – 1.5

Introduction

In data analysis, the **z-score** is a statistical measure that indicates how far a data value is from the mean in terms of standard deviations. Businesses use z-scores to identify unusual customer behavior, such as exceptionally high or low spending. This helps in customer segmentation, fraud detection, and marketing strategies.

Given:

- Mean (μ) = ₹2,000
- Standard Deviation (σ) = ₹400
- Customer Spending (X) = ₹3,200

1. Calculate the customer's z-score

Formula $Z = \frac{X - \mu}{\sigma}$: $Z = \frac{3200 - 2000}{400}$

$$Z = \frac{1200}{400} = 3$$

Answer: Z-score = 3

2. Is the customer's spending unusually high?

A z-score tells how many standard deviations a value is from the mean.

- $z = 0 \rightarrow$ Average spending
- $z = \pm 1 \rightarrow$ Slightly different
- $z = \pm 2 \rightarrow$ Unusual
- $z = \pm 3 \rightarrow$ Very unusual

Since the customer has $z = 3$, the spending is **3 standard deviations above the average**.

Answer:

Yes. The customer is an **unusually high spender** and may belong to premium customers.

3. What would a negative z-score mean?

A negative z-score indicates that the customer's spending is **below the average**.

Example:

If $z = -1$

Customer spends one standard deviation less than the average.

Answer:

A negative z-score means the customer spends **less than the average amount**.

4. If another customer has $z = -1.5$, what does it tell you?

$z = -1.5$ means

Customer spending is **1.5 standard deviations below the average**.

Actual spending: $X = \mu + z\sigma$

$$= 2000 + (-1.5)(400)$$

$$= 2000 - 600 = ₹1400$$

Answer:

The customer spends **₹1,400**, which is **below average** but not extremely unusual.

Conclusion

- Customer 1 is a very high spender.
- Customer 2 spends below average.
- Z-score helps compare values from different datasets.

2. Delivery Time Analysis

Aim

To analyze delivery time data using measures of central tendency, quartiles, interquartile range (IQR), and outlier detection for evaluating delivery performance.

Problem Statement

A delivery company records the delivery times (in minutes) for 10 deliveries. Calculate the mean, median, quartiles, IQR, identify outliers using the $1.5 \times$ IQR rule, and determine whether the mean or median better represents the typical delivery time.

Introduction

Delivery time analysis is essential for evaluating logistics and customer satisfaction. Statistical measures such as the **mean, median, quartiles, and interquartile range (IQR)** help identify the typical delivery time and detect unusually long or short deliveries. Outlier detection improves operational decision-making.

Given Data

25, 27, 28, 29, 30, 30, 31, 32, 34, 60

1. Mean and Median

Mean

Sum

$$25+27+28+29+30+30+31+32+34+60=326$$

$$\text{Mean}=326/10=32.6$$

Median

Middle values:

30 and 30

$$30+30/2=30$$

Answer

Mean = **32.6 minutes**

Median = **30 minutes**

2. Find Q1 and Q3

Lower half:

25, 27, 28, 29, 30

$$Q1 = 28$$

Upper half:

30, 31, 32, 34, 60

$$Q3 = 32$$

3. Calculate IQR

$$IQR = Q3 - Q1$$

$$= 32 - 28 = 4$$

Answer

$$IQR = 4$$

4. Outlier using $1.5 \times IQR$ Rule

Lower Limit

$$28 - 1.5(4) = 22$$

Upper Limit

$$32 + 1.5(4) = 38$$

Any value above **38** is an outlier.

Only **60** exceeds 38.

Answer

60 minutes is an outlier.

5. Which is better: Mean or Median?

Because of the outlier (60), the mean increases to 32.6.

Median remains 30.

Answer

Median better represents the typical delivery time because it is not affected by extreme values.

Conclusion

Median is preferred whenever outliers exist.

3. Two Cities' Temperatures

Aim

To compare the average temperatures and variability of two cities using the mean, variance, and standard deviation.

Problem Statement

The average daily temperatures of City A and City B are recorded for 7 days. Calculate the mean and variance for each city, compare their temperature variability, explain how two datasets can have the same mean but different standard deviations, and identify which city has more predictable temperatures.

Introduction

Weather data analysis helps compare climate conditions between different locations. Statistical measures such as the **mean, variance, and standard deviation** are used to understand average temperatures and their variability. Comparing these measures helps determine which city experiences more stable weather conditions.

City A

28, 29, 30, 30, 31, 30, 32

City B

20, 25, 30, 35, 30, 25, 45

1. Mean Temperature

City A

Sum = 210

$210/7=30$ degree C

City B

Sum = 210

$210/7=30$ degree C

Answer

Both cities have the **same mean = 30°C**.

2. Variance

City A

Squared deviations:

4,1,0,0,1,0,4

Total =10

Variance: $10/7=1.43$

City B

Squared deviations:

100,25,0,25,0,25,225

Total =400

Variance: $400/7=57.14$

3. Which city has greater variability?

City A Variance = **1.43**

City B Variance = **57.14**

Answer

City B has much greater temperature variation.

4. Why same mean but different standard deviations?

Mean only measures the average.

Standard deviation measures how spread out the values are.

City A temperatures stay close to 30°C.

City B temperatures vary from 20°C to 45°C.

5. Which city has more predictable temperatures?

City A has very low variation.

Therefore its temperatures are much more predictable.

Conclusion

- Mean alone cannot describe variability.
- Standard deviation provides additional information about consistency.

4.Data Center CPU Usage

Aim

To analyze CPU usage data using descriptive statistics and identify outliers and skewness in system performance.

Problem Statement

A data scientist records CPU usage percentages from a data center. Calculate the mean, median, mode, range, quartiles, and IQR, identify any outliers using the

1.5 × IQR rule, and determine whether the data is positively or negatively skewed.

Introduction

CPU usage is an important performance indicator in data centers. Statistical analysis of CPU usage helps monitor system performance, identify abnormal usage patterns, detect outliers, and understand the distribution of resource utilization. These analyses assist in improving system reliability and performance.

Data

40, 42, 45, 45, 46, 48, 50, 52, 55, 90

1. Mean, Median and Mode

Sum

513

Mean

$513/10=51.3$

Median: $46+48/2=47$

Mode

45

Answer

Mean = **51.3**

Median = **47**

Mode = **45**

2. Range

$$90-40=50$$

$$\text{Range} = \mathbf{50}$$

3. Find Q1, Q3 and IQR

Lower half

40,42,45,45,46

$$Q1 = 45$$

Upper half

48,50,52,55,90

$$Q3 = 52$$

IQR

$$52-45=7$$

4. Identify Outliers

Lower Limit

$$45-1.5(7)=34.5$$

Upper Limit

$$52+1.5(7)=62.5$$

$$90 > 62.5$$

Answer

90 is an outlier.

5. Determine Skewness

Mean > Median

$$51.3 > 47$$

Therefore distribution is **positively (right) skewed**.

Conclusion

The high CPU usage value (90%) increases the mean and causes right skewness.

5.Comparing Two Datasets

Aim

To compare two datasets using measures of central tendency and dispersion in order to evaluate consistency.

Problem Statement

Two data analysts are given two datasets. Calculate the mean, variance, and standard deviation of both datasets, compare their spread, determine which dataset is more consistent, and explain why the mean alone is insufficient for comparing datasets.

Introduction

Comparing datasets requires more than calculating the average. Measures such as **variance and standard deviation** describe the spread of data and help evaluate consistency. Two datasets may have the same mean but differ significantly in variability, making dispersion measures essential for proper comparison.

Dataset A

10,20,30,40,50

Dataset B

28,29,30,31,32

1. Mean

Dataset A: $150/5=30$

Dataset B: $150/5=30$

Both means = **30**

2. Variance

Dataset A

Squared deviations

400,100,0,100,400

Total =1000

Variance: $1000/5=200$

Dataset B

Squared deviations

4,1,0,1,4

Total =10

Variance: $10/5=2$

3. Standard Deviation

Dataset A: Square root of 200=14.14

Dataset B :

$\text{sqrt}2=1.41$

4. Which dataset has greater spread?

Dataset A:

SD = 14.14

Dataset B:

SD = **1.41**

Answer

Dataset A has much greater spread.

5. Which model is more consistent?

Lower standard deviation means higher consistency.

Dataset B has SD = 1.41.

Answer

Dataset B is more consistent.

6. Why is mean alone not enough?

Both datasets have the same mean (30), but their values are distributed very differently.

- Dataset A ranges widely from **10 to 50**, indicating high variability.
- Dataset B stays close to **30**, indicating low variability.

Therefore, relying only on the mean can be misleading. Measures such as **variance** and **standard deviation** are needed to understand the spread and consistency of the data.